

Instruction for Kazakh Morphological Segmentation

(CSE-based sentence segmentation using `chunk_099.txt`)

1. Prepare the Google Colab environment

1. Open a new notebook in **Google Colab**.
 2. Go to **Runtime** → **Change runtime type**.
 3. Select **Hardware accelerator: GPU** (optional, but recommended).
 4. Click **Save**.
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2. Mount Google Drive and set the working directory

Run the following cell to mount Google Drive and move to the project directory:

```
from google.colab import drive
drive.mount('/content/drive')

curr_dir = "/content/drive/MyDrive/SEGMENTATION_QAZ"
%cd "$curr_dir"
```

Make sure that the directory `SEGMENTATION_QAZ` contains all required input files (see Section 3).

3. Install required libraries

In a separate cell, install the required Python packages:

```
!pip install xlwt
!pip install sacrebleu
```

The remaining libraries (`re`, `os`, `numpy`, `openpyxl`, `tqdm`, etc.) are part of the standard Python/Colab environment.

4. Prepare input resources

Ensure the following files are present in the working directory:

File name	Description
<code>chunk_099.txt</code>	Input Kazakh text (UTF-8, one sentence per line)
<code>qaz_stems_unik_edit.xlsx</code>	List of Kazakh stems
<code>qaz_endings_seg.xls</code>	Endings and their CSE-based segmentation
<code>stop_words.txt</code>	Stop-word list (one word per line)

Note:

`chunk_099.txt` is a representative subset of the full corpus (284,707 sentences) used in the

paper.
The same code was applied to the full dataset.

5. Configure input and output paths

In the script, verify the following parameters:

```
input_path = "chunk_099.txt"
output_dir = "segmented_chunk_099"
chunk_size = 5000
```

- `input_path` — input text file
 - `output_dir` — directory where segmented chunks will be saved
 - `chunk_size` — number of sentences processed per chunk
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6. Run the segmentation script

1. Copy the full segmentation code into **one Colab cell**.
2. Run the cell.

The program will:

- read the input text,
- split it into chunks of 5,000 sentences,
- perform CSE-based morphological segmentation,
- save the results incrementally.

Progress is displayed using `tqdm`.

7. Output structure

After execution, the output directory will have the following structure:

```
segmented_chunk_099/
├── chunk_000.txt
├── chunk_001.txt
├── chunk_002.txt
└── ...
```

If we split chunks of less than 5,000 sentences, because `chunk_099` has 5,000 sentences.

Each file contains morphologically segmented sentences, with morpheme boundaries marked using @@.

Example output line:

```
жаса@@ й@@ ды адам@@ дар
```

8. Notes on reproducibility

- The segmentation logic is deterministic and rule-based (CSE model).
- The same script was used for:
 - intrinsic tokenizer evaluation,
 - training FEMSeg_kaz neural models,
 - external and extrinsic experiments.
- Researchers may replace `chunk_099.txt` with any independently collected Kazakh corpus following the same format.

9. License and usage

- **Software:** MIT License
- **Data:** CC BY 4.0
- Intended for **research and reproducibility purposes.**